

Scalability & Future Plan

Date	29 June 2026
Team ID	N/A
Project Name	OptiCrop Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	No cloud deployment	Medium	High
2	No database	Medium	Medium
3	Single User Usage	Low	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single User	Multi User support
2	Data Storage	Local	Cloud database
3	Performance	Local processing	Optimized cloud Processing
4	Security	Basic validation	Authentication and Encryption

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 1	Cloud deployment	3 Months	Improved accessibility, reduced infrastructure cost, scalable environment
Phase 2	Database Integration	6 Months	Centralized data management, faster queries, better reliability
Phase 3	Mobile Application	9 Months	Wider reach, user convenience, increased engagement and adoption